

Introduction to Professional Practices in IT

CS4001 – Professional Practices in IT

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Week 10 - Intellectual Property Rights

Ownership

- As a general rule, the copyright in a work belongs initially to its author. If the work is jointly written by several authors, they jointly own the copyright.
- If the author is an employee and has written the work as part of his or her job, then the copyright belongs to the employer, unless there is an explicit, written agreement to the contrary.
- The employer owns the copyright only if the author is legally an employee. If the author is an independent contractor, he or she will own the copyright unless there is an agreement to the contrary. For this reason, if a company commissions an independent contractor (freelance programmer) to write software, it is important to have a formal agreement regarding ownership of the copyright in the resulting software.

Licensing

- It is very common for the owner of the copyright in a piece of software to license other people or organizations to carry out some of the activities that are otherwise the exclusive right of the copyright owner.
- The copyright remains the property of the owner, but the *licensees* (the people to whom the software is licensed) acquire certain rights.

PATENTS

- A patent is a temporary right, granted by the state, enabling an inventor to prevent other people from exploiting his invention without his permission.
- Unlike copyright, *it does not come into existence automatically; the inventor must apply for the patent to be granted.* However, the protection it gives is much stronger than copyright, because the grant of a patent allows the person owning it (the *patentee*) to prevent anyone else from exploiting the invention, even if they have discovered it for themselves.

What can be patented?

In Europe, the law relating to patents is based on the European Patent Convention. This was signed in 1973 by 27 European countries and came into force in 1978. The UK's obligations under the Convention were implemented in the Patents Act 1977, although there have been some subsequent modifications. The 1977 Act states that an invention can only be patented if it must meet these four conditions:

It's new: Nobody else has made or shared this idea before.

It's inventive: It's not an obvious solution that anyone could think of.

It can be used: The invention should work in real-life industries.

It's allowed: It must not fall into categories excluded from patents (like laws of nature or abstract ideas).

What Patent Act Excludes

- Following the European Patent Convention, the Patents Act 1977 excludes the following:
- **Scientific theories:** The theory of gravity cannot be patented although a machine that uses it in a novel way could be.
- **Mathematical methods:** This means, for example, that the methods used for carrying out floating point arithmetic cannot be patented. A machine that uses the ideas can however be patented.
- **A literary, dramatic, musical or artistic work or any other aesthetic creation:** As we have already seen, these are protected by copyright.
- **The presentation of information:** Again, this is covered by the law of copyright.
- **A scheme, rule or method for performing a mental act, playing a game or doing business, or a program for a computer.**

Obtaining a patent

- Unlike copyright, which comes into existence automatically when the protected work is recorded, whether in writing or otherwise, **a patent must be explicitly applied for. Applying for a patent can be an expensive and time-consuming business.**
- Patents are granted by national patent offices. Inventors who want protection in several different countries must, in principle, apply separately to the patent offices of each country.
- The full patent specification needs to be prepared by a specialist patent attorney, and it can take up to four years for the process of obtaining patents to be completed.
- In Pakistan **Governing Body: Intellectual Property Organization of Pakistan (IPO-Pakistan)**

Enforcing a patent

- **Patents don't guarantee easy protection:** Even if you have a patent, stopping others from using your idea may require legal action.
- **Main issue:** 'Prior art': A common challenge is proving that your invention is unique and wasn't already known before.
- **Legal battles are costly:** Defending your patent or fighting against someone else's patent takes a lot of time and money.
- The problem is that enforcing a patent that you own or challenging a patent held by someone else is a time-consuming and expensive process.

Software Patents

- A software patent does not cover the software's code itself (this is typically protected by copyright) but rather the underlying process, system, or method.
- For example, a new algorithm for improving image processing speed in medical imaging software could potentially be patented if it's a novel, non-obvious, and technically complex solution.

Software Patents (Cont.)

- The European Patent Office has been granting patents for software since 1998, as has the UK Patent Office.
- Patent offices in the different European countries have adopted different policies towards the patenting of software, with the result that there is much confusion about what is and what is not patentable.
- The result is that there is a conflict between the law and practice, a very undesirable situation.

CONFIDENTIAL INFORMATION

Information that is shared in trust and must not be revealed or misused.

It cannot be “stolen” in a legal sense, but using or disclosing it without permission can lead to civil legal action.

Example: A software company’s source code, pricing model, or client list.

Obligation of Confidence

- Exists when information is given under circumstances where a **reasonable person** would understand it's **meant to stay private**.
- Can arise from:
 - A **contract clause** (e.g., in consultancy or software development agreements).
 - A **Non-Disclosure Agreement (NDA)** — a standalone contract that ensures confidentiality.
- Example: Two IT firms discuss a potential partnership. Before sharing technical details, they sign an NDA so neither can later use the other's information unfairly.

Confidentiality & Patents

- Ideas intended for **patent protection** must **not be made public** before filing.
- Sharing such information **without confidentiality protection** can lead to the **loss of patent rights**.
- *Example:* An inventor discusses a new AI algorithm with another company **without an NDA**, and the idea leaks, now they **can't patent it**.

Skill vs Confidentiality

- **Skills learned** (e.g., programming in Python, using design tools) belong to the **employee**.
- **Confidential info** (e.g., product plans, client data) belongs to the **employer**.
- You can **use your skills** in your next job, but **not the company's private strategies**.

Whistleblowing

- **Whistleblowing** happens when an **employee exposes wrongdoing** (like illegal, unethical, or dangerous activities) inside their organization, even though their employment contract or NDA says they must keep company information confidential.
- So, it's when someone "blows the whistle" on their own company because they believe **public safety, honesty, or justice** is more important than loyalty to their employer.

Whistleblowing & Public Interest

- Courts may allow disclosure if it's in the public interest (e.g. exposing fraud or safety violations). In such cases, the employee might **not be legally punished**.
- However, the employee may still lose their job even if not legally punished.
- Example: An employee leaks proof of **toxic waste dumping** by a company. The court may protect them legally, but the company might still dismiss them.

Public Interest Disclosure Act.

- In 1998, the UK Parliament passed the Public Interest Disclosure Act. which provides some protection for employees in these circumstances.
- First, the Act defines what sort of disclosure of information is covered. A **qualifying disclosure** that is, a disclosure to which the Act applies means any disclosure of information that the person making the disclosure.

Public Interest Disclosure Act.

- Disclosure act reasonably believes shows that one or more of the following has occurred or is about to occur:
 - a criminal offence;
 - failure to comply with a legal obligation;
 - a miscarriage of justice;
 - danger to health and safety;
 - environmental damage;
 - information showing that any of these has been concealed
- A worker making a qualifying disclosure will only be protected against victimization if the disclosure is made in the right circumstances. In this case, the disclosure is known as a protected disclosure

TRADEMARKS AND PASSING OFF

- The law regarding trademarks in the UK is based on the Trademarks Act 1994, which consolidated and updated existing legislation. The Act defines a trademark as:
- ***Any sign capable of being represented graphically which is capable of distinguishing goods or services of one undertaking from those of other undertakings. A trademark may consist of words (including personal names), designs, letters, numerals or the shape of goods or their packaging.***
- Even where a trademark is not registered, action can be taken in the civil courts against products that imitate the appearance or 'get up' of an existing product. This is known as the tort of 'passing off'. Tort(legal wrong)

TRADEMARKS AND PASSING OFF

- Provision is made for registering trademarks and most trademarks are now registered. In the UK this is done through the UKIPO. It maintains a database of registered trademarks and their owners that can be searched over the internet.
- Trademarks are an effective way of protecting retail packaged software from piracy.

TRADEMARKS

There are rules about what can be registered as a trademark. For example, place names or people's names usually can't be trademarked.

The 1994 Act makes it illegal to:

- Use a trademark that you don't own or have permission to use on goods.
- Sell or offer goods or packaging with an unauthorized trademark.
- Import or export goods with an unauthorized trademark.
- Have goods or packaging with an unauthorized trademark for sale or hire in business.

Domain Names

- Internet domain names are ultimately managed by the Internet Corporation for Assigned Names and Numbers (ICANN). ICANN is an internationally organized, non-profit making corporation. Its main responsibility is ensuring the 'universal resolvability' of internet addresses; that is, ensuring that the same domain name will always lead to the same internet location wherever it is used from and whatever the circumstances.
- In practice, ICANN delegates the responsibility for assigning individual domain names to other bodies, subject to strict rules.

Domain Names

- Domain names were originally created to make it easier for computers to connect on the internet. However, because they are easy to remember, they are now often used by **businesses to identify themselves and are commonly featured in advertising.**
- It's natural that companies want to use **their trademarks or company names as domain names.** However, this can lead to conflicts between trademarks and domain names, as the two systems work differently.
- Trademarks are registered with authorities in specific countries or regions, giving the owner rights to the trademark within that area. Different people can own the same trademark for different types of products.

Domain Names

The owner of the trademark acquires rights over the use of the trademark in a specific country or region. Identical trademarks may be owned by different persons in respect of different categories of product.

Domain Names

"Dove" – Two Companies, Different Products

- **Dove (Soap and Personal Care)**
 - Owned by: **Unilever**
 - Products: Soaps, shampoos, lotions, skincare
 - Trademarked in: **Personal care category** in most countries
- **Dove (Chocolate and Confectionery)**
 - Owned by: **Mars Inc.**
 - Products: Chocolate bars, ice cream, sweets
 - Trademarked in: **Food and confectionery category**

Why is this allowed?

- Trademark law is territorial and category-specific.
- If the products are in unrelated markets and consumers are unlikely to be confused, identical marks can legally coexist.

Cyber squatting

- The inconsistencies between the two different systems of registration has made it possible for people to register, as their own domain names, trademarks belonging to other companies. This is sometimes known as cyber squatting. They then offer to sell these domain names to the owner of the trademark at an inflated price.
- It is usually cheaper and quicker for the trademark owner to pay up than to pursue legal remedies, even when these are available.
- Example: Nissan.com

WIPO

- **WIPO** stands for the **World Intellectual Property Organization**.
- Established in **1967**, based in **Geneva, Switzerland**.
- Its mission is to promote the **protection of intellectual property (IP)** globally.

Functions of WIPO

- Develops International IP Laws
- Dispute Resolution (UDRP)
- Supports Innovation & IP Systems
- Global IP Services
- IP Databases & Tools

WIPO ensures **global cooperation** in protecting IP rights.

Simplifies **cross-border filing** and enforcement.

Plays a key role in **digital and internet-era issues** like **cybersquatting**.

WIPO report

- In 1999, the WIPO published a report entitled **'The management of internet names and addresses: Intellectual property issues'** The report recommended that ICANN adopt a policy called the Uniform Domain Name Dispute Resolution Policy (UDRP), which includes specific provisions against cyber squatting. This policy has proved reasonably effective. Within two years, over 3,000 complaints had been dealt with by one of the arbitration centres alone, with 80 per cent being resolved.

Examples - Tesla.com Case

Background: Before **Tesla Motors** became a global brand, the domain **Tesla.com** was already owned by a **cybersquatter named Stuart Grossman**. Tesla had to operate under **TeslaMotors.com** for years.

Issue: **Grossman** had registered Tesla.com well before the electric car company became prominent. Despite having **no product, company, or public-facing content**, the domain owner **refused to sell** for a reasonable price.

Resolution: In **2016**, Tesla finally **acquired Tesla.com** — reportedly for a **multi-million-dollar sum**, though the exact amount was **undisclosed**. This allowed Tesla to simplify its brand and web presence from **TeslaMotors.com → Tesla.com**.

Examples - MikeRoweSoft.com Case (Microsoft vs. Mike Rowe)

Background: In **2003**, Canadian teenager **Mike Rowe** (17 years old) registered the domain **MikeRoweSoft.com** as a pun on his name. His intent was reportedly to **create a small web design business**.

Issue: **Microsoft** claimed the domain infringed on its trademark because it **sounded phonetically identical** to “Microsoft”. Microsoft **sent a cease-and-desist letter**, offering Rowe \$10 for the domain matching the registration cost.

Resolution:

The two parties **settled amicably**:

- Rowe transferred the domain to Microsoft.
- In return, he received:
 - An **Xbox console and games**
 - A **trip to Microsoft headquarters**
 - Some **training materials** and a **certification course**
- Rowe also auctioned the legal documents online to raise funds for his education.

Case Study: The Rise and Risks of NeuroTech AI Ltd

NeuroTech AI Ltd., a UK-based startup, has developed a revolutionary wearable neural interface, a device that reads brain signals and translates them into real-time digital commands. The technology is aimed at helping disabled individuals control digital devices using only their thoughts.

The company:

- Has **patented** its unique neural signal processing algorithm.
- Keeps its device calibration method as a **trade secret**.
- Uses a unique brand name and logo: "**NeuroPulse™**", which is **trademarked** in the UK and EU.
- Uses internally developed software with a **copyright** notice and EULA restricting its use.
- Sells to hospitals under **site licences**.

Case Study Cont: Situation 1

- Amir, a senior software engineer, discovers that during rushed development for launch, the team **copied portions of code** from a competitor's open-source library, which **required attribution and share-alike licensing**, but **none was given**.
- He also finds out that the **AI sometimes misfires**, producing **false signals**, which could lead to dangerous consequences in hospital settings. When he reports this internally, his concerns are dismissed as "technical noise" by management.

Case Study Cont: Situation 2

- While NeuroTech AI owns **neuropulse.ai**, they discover that someone has registered **neuropulse.com** and **neuropulsehealth.com** both redirecting to **ads and competitor sites**. The domains were registered **after** the company's trademark registration.

Question 1

Did NeuroTech violate copyright law by using open-source code without attribution?

Expected Response

- Yes. If the open-source code was licensed under a **share-alike or attribution license**, failing to:
- Attribute the original author,
- Share their modifications under the same licence, or
- Include a copy of the licence and copyright notice

It would constitute a **copyright infringement**.

Question 2

How is the calibration method protected under trade secret law? What precautions should the company take to ensure this protection?

Expected Response

The calibration method is protected as a **trade secret** if:

- It is **not publicly known**,
- Provides **commercial value** from being secret,
- And **reasonable steps** have been taken to keep it confidential.

Precautions include:

- Non-disclosure agreements (NDAs),
- Internal access controls (e.g. limited employee access),
- Training on confidentiality

Question 3

Under the Public Interest Disclosure Act 1998, is Amir's report of both copyright infringement and product safety issues considered protected disclosure? Why or why not?

Expected Response

Yes, potentially a **protected disclosure**, because:

- He reasonably believes there's a **breach of legal obligation** (copyright infringement),
- And **danger to public health/safety** (AI misfires).
- He raised it **internally first**, which follows recommended procedure.
- His intent seems aligned with the **public interest**, not personal gain.

Question 4

Keeping this scenario in mind How can a company **proactively prevent cybersquatting** when launching a new product or brand?

Expected Response

- Register trademarks early, in all key markets
- Secure multiple domain name variants (.com, .net, country domains)
- Use trademark monitoring services or domain watchlists
- Set up a brand/domain enforcement strategy (e.g. WIPO readiness)